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HANDSET REV A / SYSTEM SCHEMATIC INDEX

Engineering draft. Retained ESP32-S3 / MakerFocus 3000 mAh / external display / all peripheral interfaces.

A

ESP32 host, boot and reset



File: core.kicad_sch

Key matrix and trackball



File: keys.kicad_sch

USB-C data and ESD



File: usb.kicad_sch

B

Fused USB input and eFuse



File: usb-input.kicad_sch

USB-C source detection and permission



File: usb-detect.kicad_sch

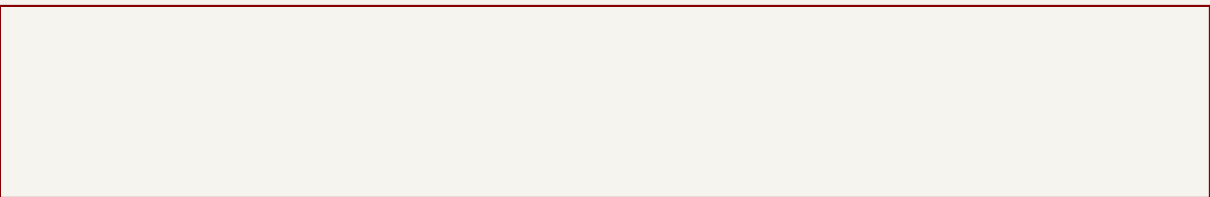
Battery, thermistor and programmable charger



File: power.kicad_sch

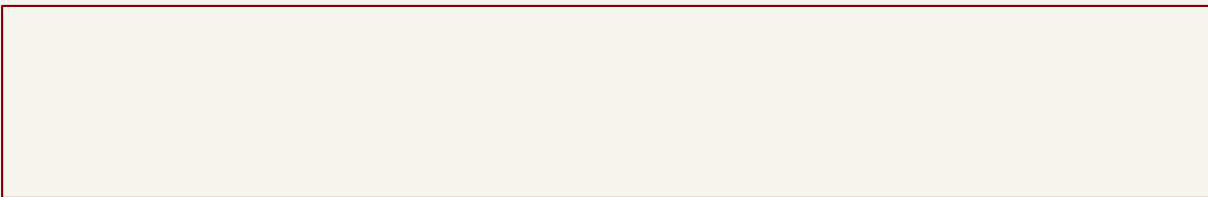
C

Power control and fuel gauge



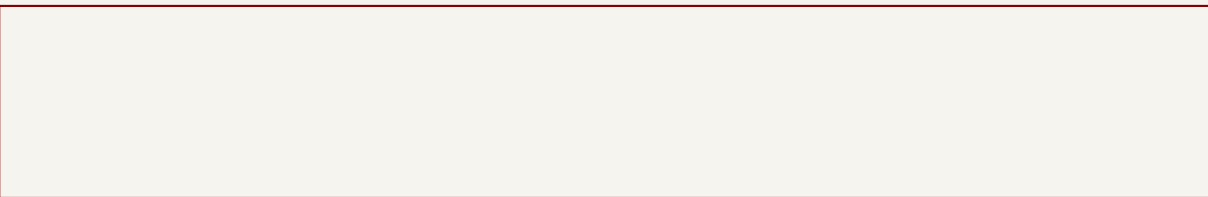
File: power-control.kicad_sch

Main 3.3 V buck-boost supply



File: reg3.kicad_sch

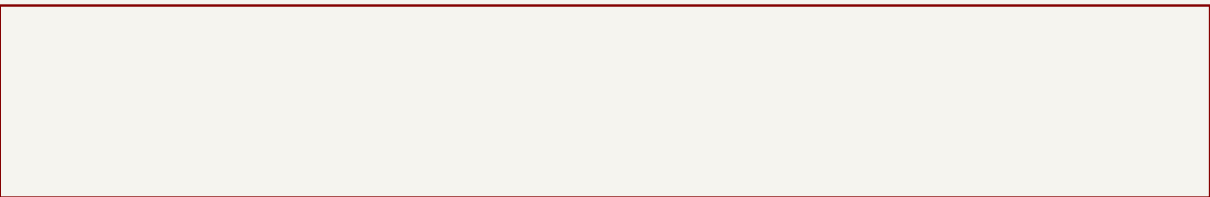
Auxiliary 5 V and 1.8 V supplies



File: aux-power.kicad_sch

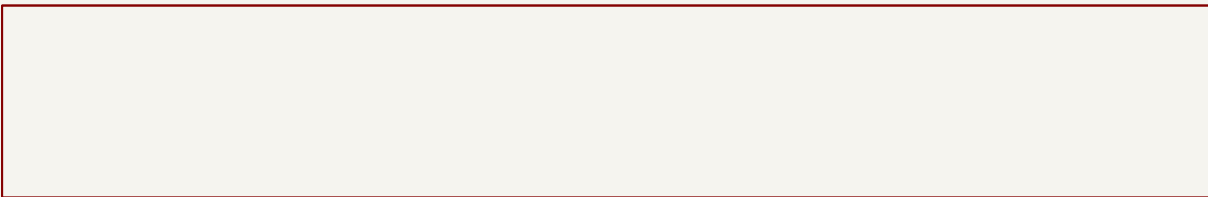
D

Regulated modem supply and enable



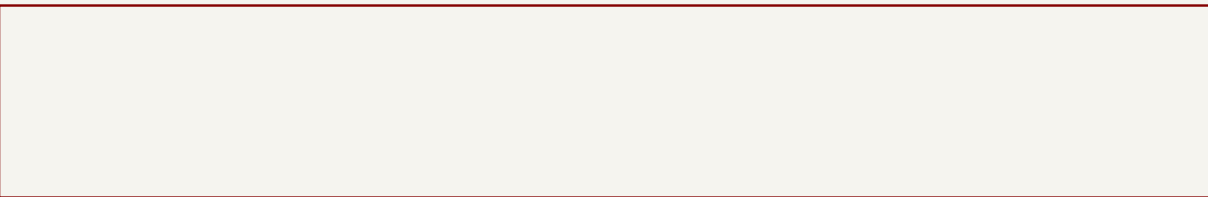
File: modem-power.kicad_sch

SX1262 LoRa radio



File: radios.kicad_sch

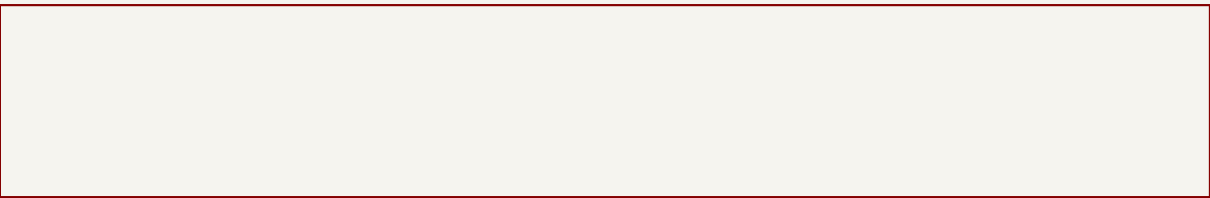
CC1101 sub-GHz radio



File: cc-radio.kicad_sch

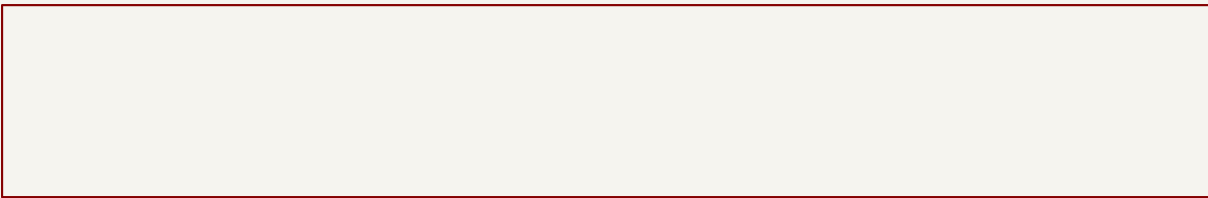
E

125 kHz reader analog circuit



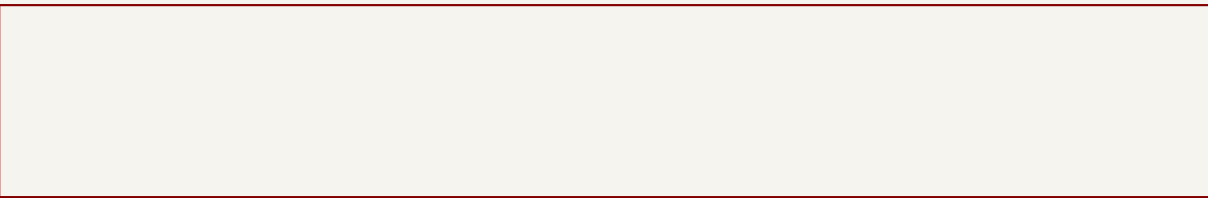
File: lf-radio.kicad_sch

LF reader logic translation



File: lf-support.kicad_sch

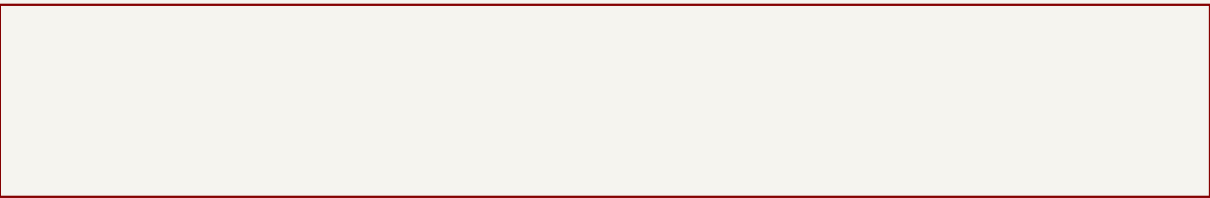
GNSS receiver and passive antenna



File: gnss-antennas.kicad_sch

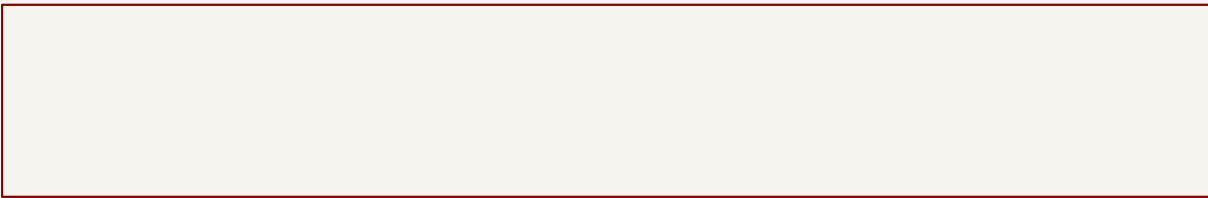
F

NFC reader, oscillator and local supplies



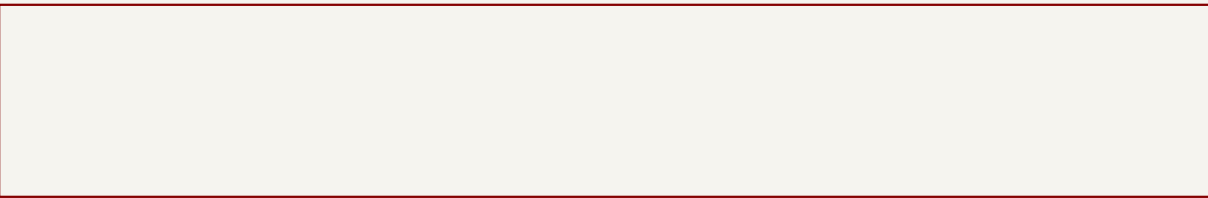
File: nfc.kicad_sch

Media speaker amplifier and digital microphone



File: audio.kicad_sch

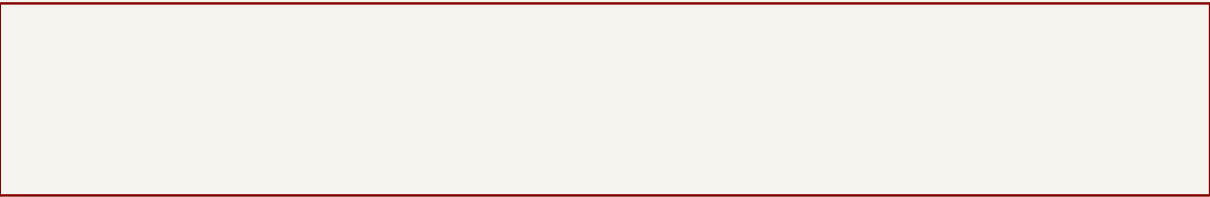
Cellular adapter and call-audio harnesses



File: interfaces.kicad_sch

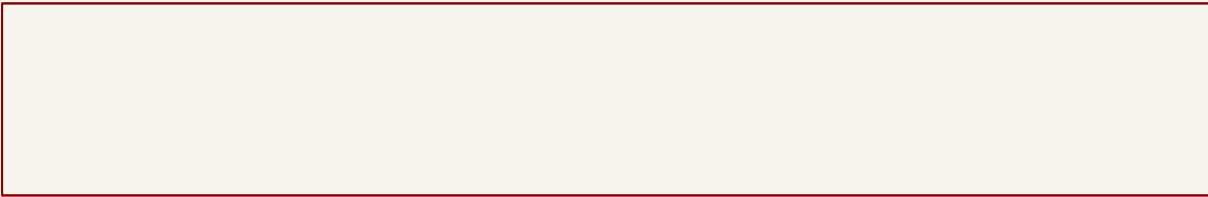
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Protected external module port



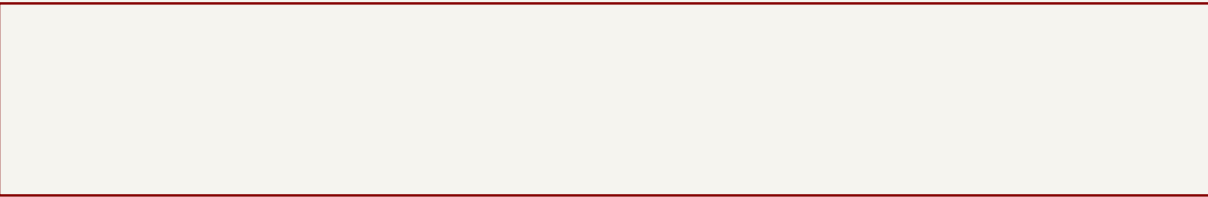
File: protection.kicad_sch

MicroSD and case-mounted Infrared



File: storage-ir.kicad_sch

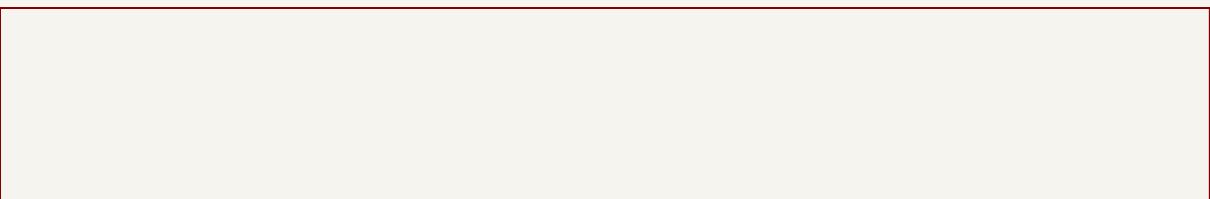
External display, touch and backlight



File: display.kicad_sch

H

Always-on charging supervisor

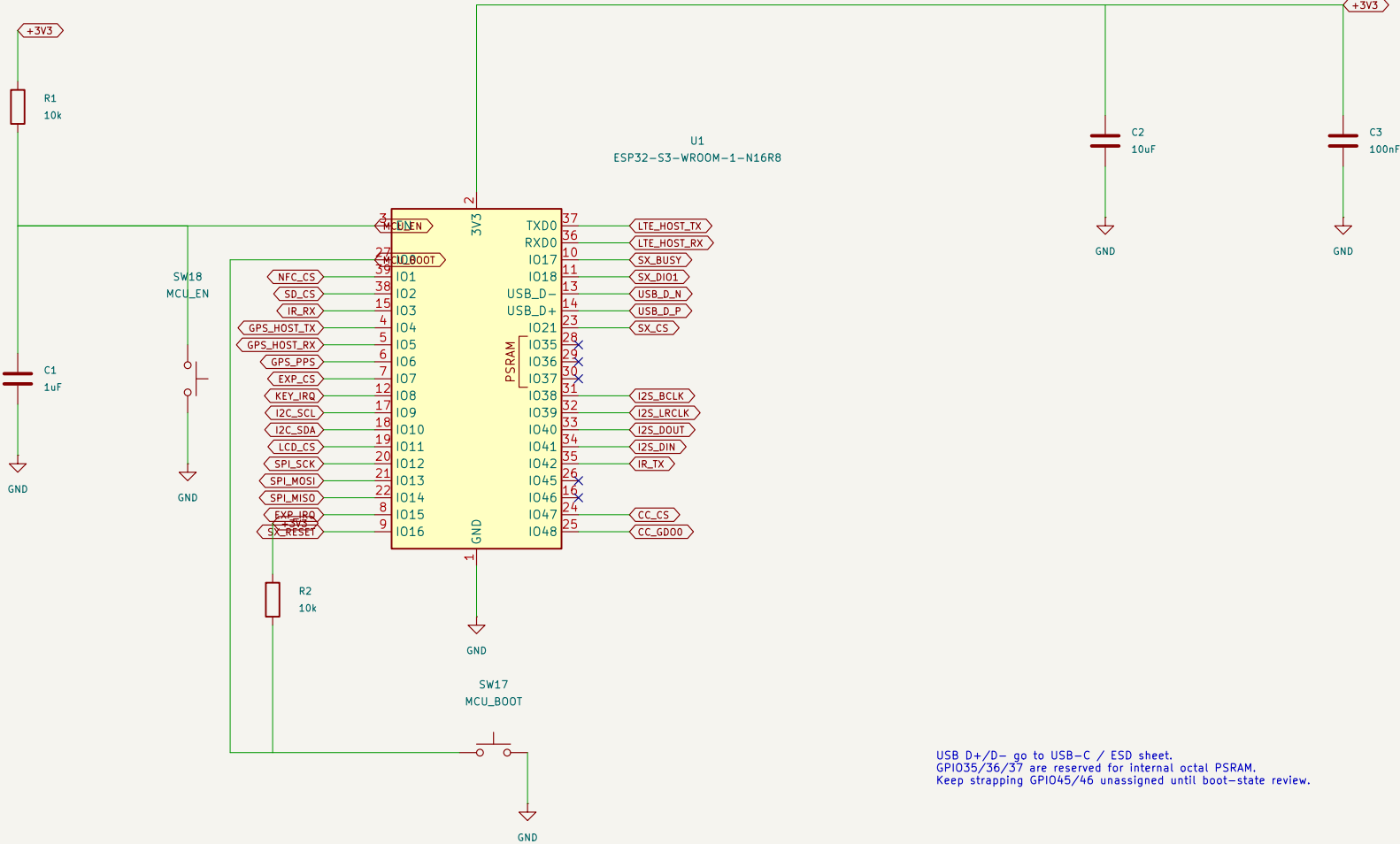


File: power-supervisor.kicad_sch

Sheet: /		
File: handset.kicad_sch		
Title:		
Size: A2	Date:	Rev:
KiCad E.D.A. 10.0.6	Id: 1/23	

ESP32 host, boot and reset

Retain N16R8. Module pads 28/29/30 belong to octal PSRAM. USB programming uses the USB sheet.



USB D+/D- go to USB-C / ESD sheet.
GPIO35/36/37 are reserved for internal octal PSRAM.
Keep strapping GPIO45/46 unassigned until boot-state review.

RESET / EN: 10k pull-up + 1uF
BOOT held low during reset selects download mode.

Circuit review in progress – not released for fabrication

Sheet: /ESP32 host, boot and reset/
File: core.kicad_sch

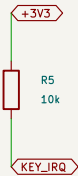
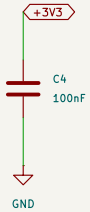
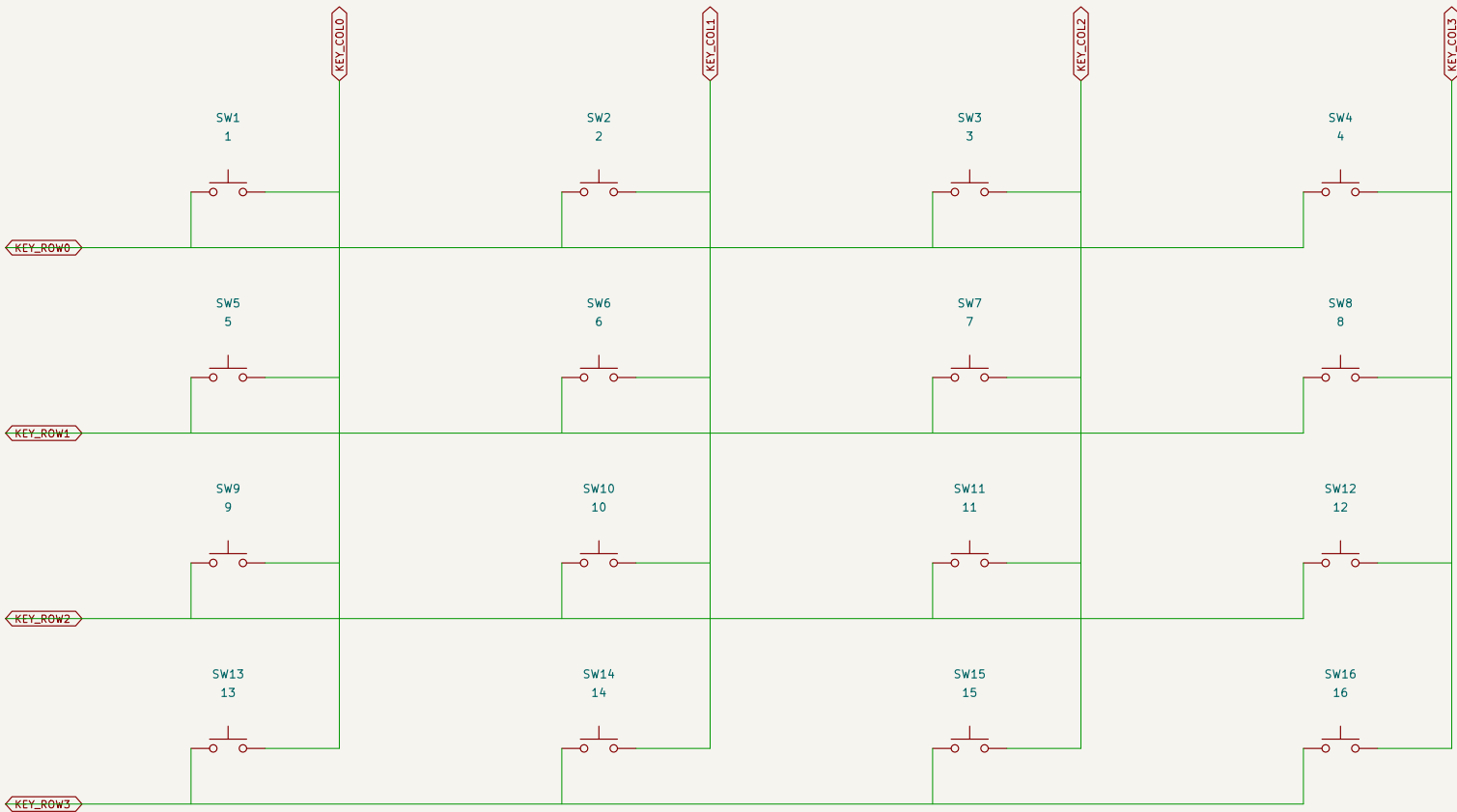
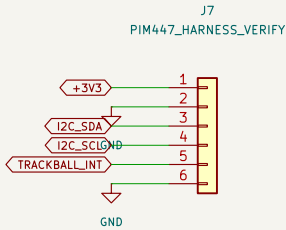
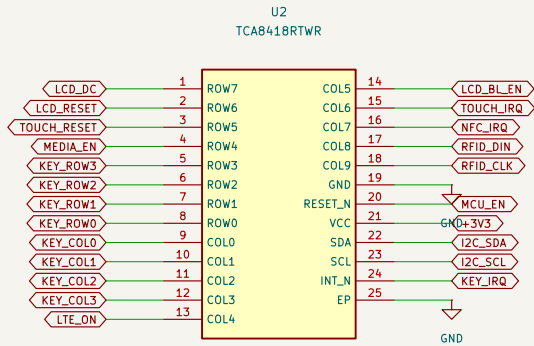
Title: ESP32 host, boot and reset

Size: A3
Date: 2026-09-24

Rev: A – engineering draft
Id: 2/23

Key matrix and trackball

TCAB418: 4 rows x 4 columns; remaining pins are firmware-configured GPIO. Confirm PIM447 harness orientation.



Circuit review in progress – not released for fabrication

Sheet: /Key matrix and trackball/
File: keys.kicad_sch

Title: Key matrix and trackball

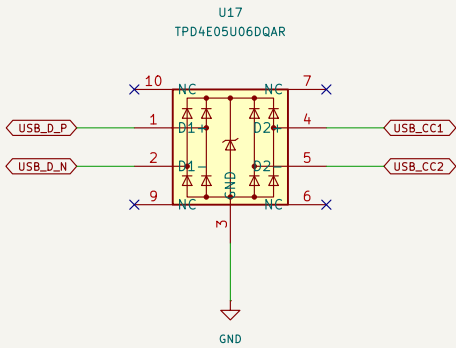
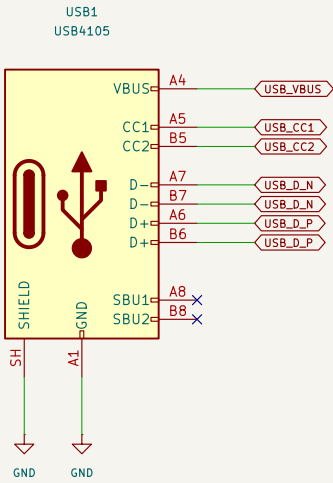
Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft
Id: 3/23

USB-C data and ESD

CC1/CC2 terminate at the sink controller on usb-detect. Data pair goes to the ESP32. Shield/ground return needs layout review.



Circuit review in progress – not released for fabrication

Sheet: /USB-C data and ESD/
File: usb.kicad_sch

Title: USB-C data and ESD

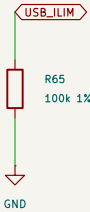
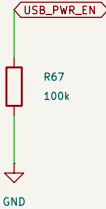
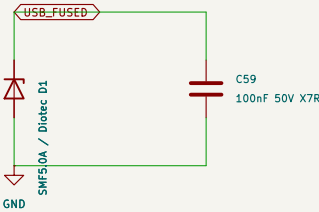
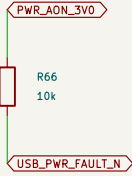
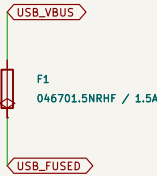
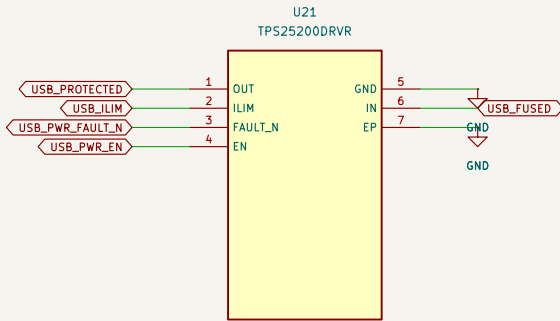
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KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft
Id: 4/23

Fused USB input and eFuse

TPS25200 R_ILIM = 100k, 1%. F1/TVS pulse coordination requires bench validation. No USB PD negotiation.



Supply flag: external USB VBUS through F1. No internal source is implied.

Circuit review in progress – not released for fabrication

Sheet: /Fused USB input and eFuse/
File: usb-input.kicad_sch

Title: **Fused USB input and eFuse**

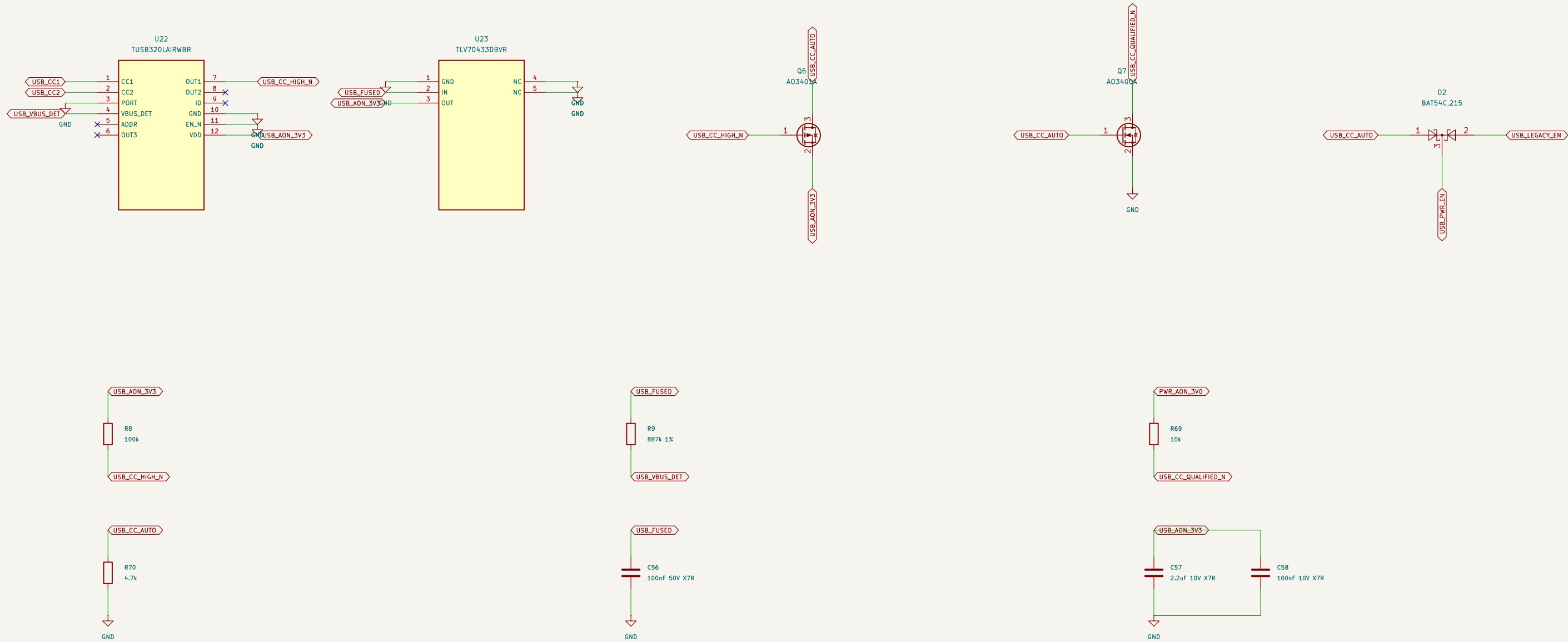
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KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: **A – engineering draft**
Id: 5/23

USB-C source detection and permission

Sink-only internal Rd. OUT1 low authorizes Type-C 1.5A/3A. Legacy permission requires USB enumeration; see charger-policy.json.



Battery, thermistor and programmable charger

Verify actual battery polarity. J2 is a battery-mounted 10k B3435 NTC. CE defaults OFF. Firmware must apply charger-policy.json.

Charge enable: default OFF
CHG_ENABLE alone cannot charge; U25 CHG_ARM is also required.

Battery-mounted thermistor
Do not substitute a fixed resistor.

Protected 103665 pack / 3000 mAh
F2 does not enforce the pack operating-current limit.

Q5 AND Q8 must both conduct to enable charging.
U25 PA0 reset/high-Z -> R79 pulls Q8 OFF.
Preload PA0 low; arm only after verified charger policy.
IWDG reset and GPIO discharge timing require bench qualification.
Supply flags identify the external protected pack through F2 and its return.

Circuit review in progress – not released for fabrication

Sheet: /Battery, thermistor and programmable charger/
File: power.kicad_sch

Title: Battery, thermistor and programmable charger

Size: A3	Date: 2026-09-24	Rev: A – engineering
KiCad E.D.A. 10.0.6		Id: 7/23

Engineering draft / named nets are common / NC = intentional unused pin

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Circuit review in progress – not released for fabrication

Sheet: /Battery, thermistor and programmable charger/
File: power.kicad_sch

Title: Battery, thermistor and programmable charger

Size: A3	Date: 2026-09-24	Rev: A – engineering
KiCad E.D.A. 10.0.6		Id: 7/23

Engineering draft / named nets are common / NC = intentional unused pin

Size: A3	Date: 2026-09-24	Rev: A – engineering draft
KiCad E.D.A. 10.0.6		Id: 7/23

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Sheet: /Battery, thermistor and programmable charger/
File: power.kicad_sch

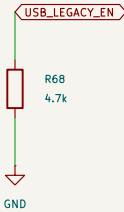
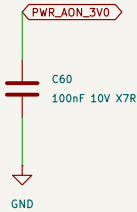
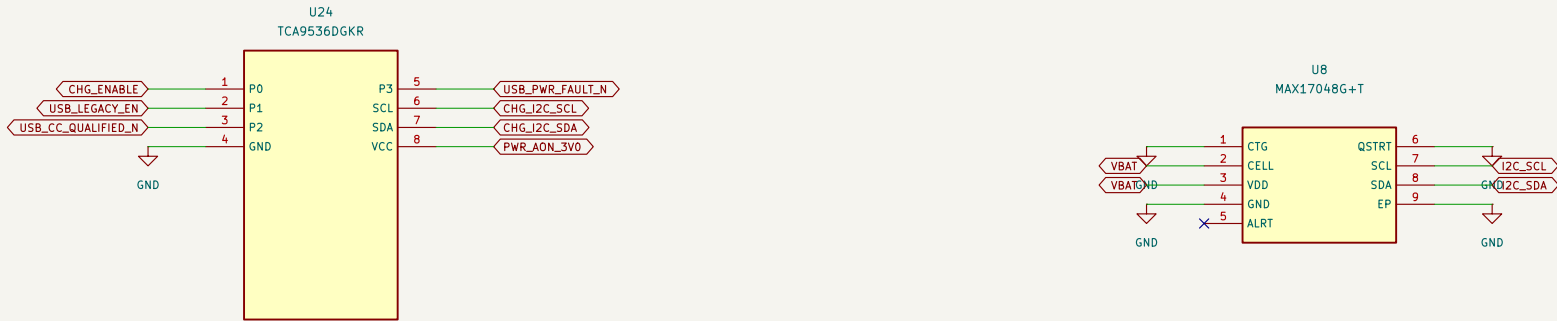
Title: Battery, thermistor and programmable charger

Size: A3	Date: 2026-09-24	Rev: A – engineering
KiCad E.D.A. 10.0.6		Id: 7/23

Engineering draft / named nets are common / NC = intentional unused pin

Power control and fuel gauge

I2C: TCA9536 0x41, MAX17048 0x36, charger 0x6A. Preload low output latches before enabling charger—control outputs.



Circuit review in progress – not released for fabrication

Sheet: /Power control and fuel gauge/
File: power-control.kicad_sch

Title: Power control and fuel gauge

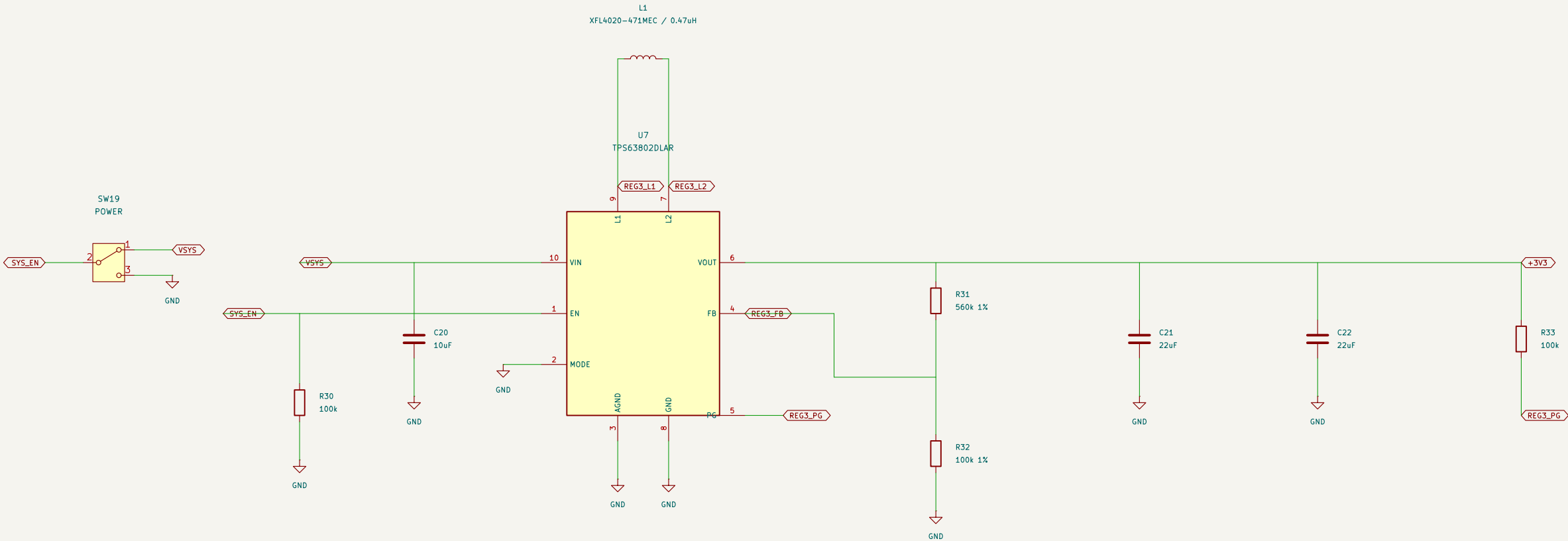
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KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft
Id: 8/23

Main 3.3 V buck–boost supply

TPS63802 local PCB copper is routed. External distribution, effective capacitance, thermal limits and load transients remain to qualify.



0.6 V reference x (1 + 560k / 100k) = 3.3 V nominal.
Feedback senses the output capacitor node; keep it clear of the switching loop.

SW19 controls SYS_EN.
Battery / charger circuitry can remain energized while logic rails are off.

Circuit review in progress – not released for fabrication

Sheet: /Main 3.3 V buck–boost supply/
File: reg3.kicad_sch

Title: Main 3.3 V buck–boost supply

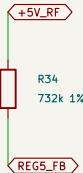
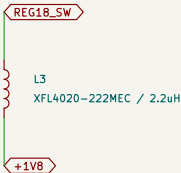
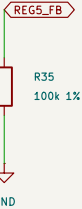
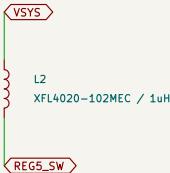
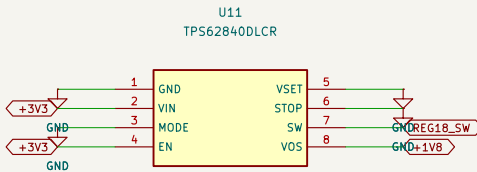
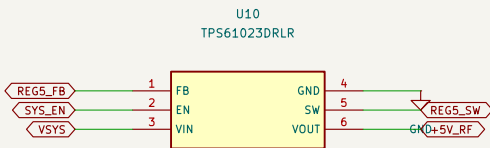
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KiCad E.D.A. 10.0.6

Date: 2026–09–24

Rev: A – engineering draft
Id: 9/23

Auxiliary 5 V and 1.8 V supplies

Switching loops must be short in PCB layout. The 5 V rail also supplies display backlight and the LF reader.



Circuit review in progress – not released for fabrication

Sheet: /Auxiliary 5 V and 1.8 V supplies/
File: aux-power.kicad_sch

Title: Auxiliary 5 V and 1.8 V supplies

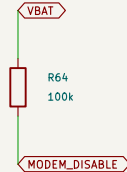
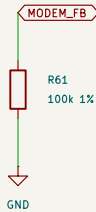
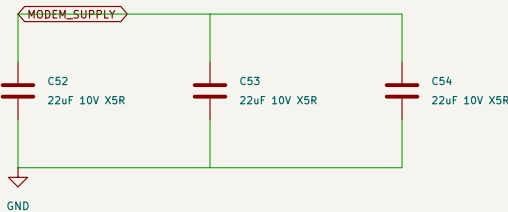
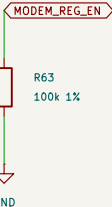
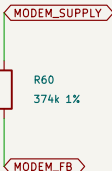
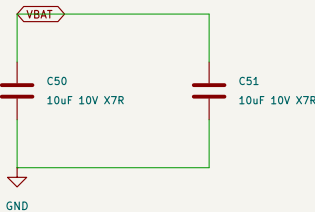
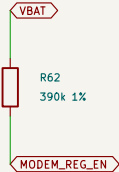
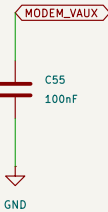
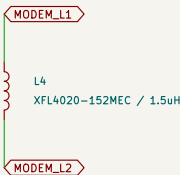
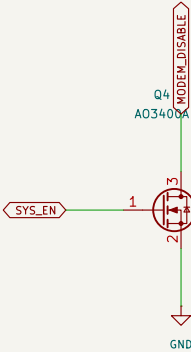
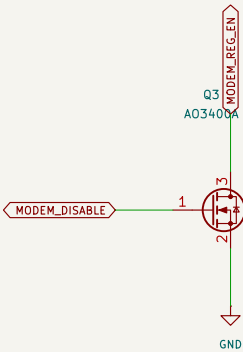
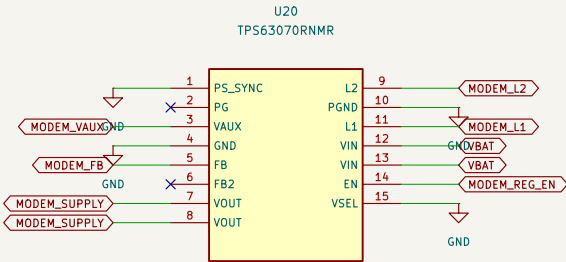
Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft
Id: 10/23

Regulated modem supply and enable

Nominal output 3.792 V. Exact modem burst current and MakerFocus pack current budget are unresolved release blockers.



Circuit review in progress – not released for fabrication

Sheet: /Regulated modem supply and enable/
File: modem-power.kicad_sch

Title: Regulated modem supply and enable

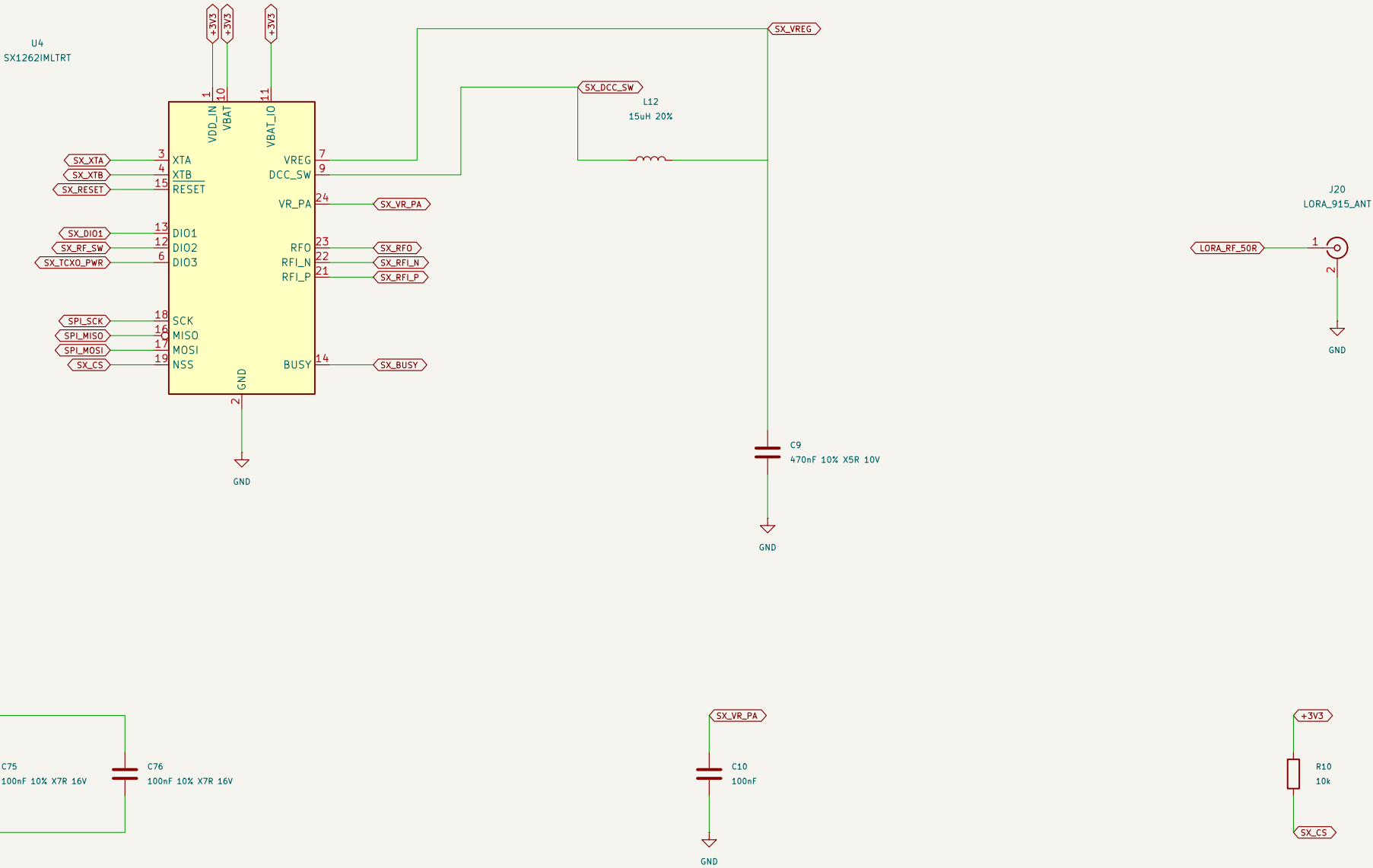
Size: A3
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Date: 2026-09-24

Rev: A – engineering draft
Id: 11/23

SX1262 LoRa radio

DC-DC support captured from Semtech DS rev1.2 Fig14-2 and Table5-3. Clock and RF frontend remain incomplete. Do not power.



C8: VBAT pin10. C75: VDD_IN pin1. C76: VBAT_IO pin11.
All use the same +3V3 rail; place each close to its pin.
C9 replaces the incorrect 100nF with Semtech reference 470nF.

SetRegulatorMode(0x01) only in STDBY_RC, after hardware qualification.
L12 supplies the core, not the SX1262 high-power PA.
TCXO, PA feed, matching/filter and RF switch are still missing.
C10 PA bypass and J20 antenna connection remain provisional.

Circuit review in progress – not released for fabrication

Sheet: /SX1262 LoRa radio/
File: radios.kicad_sch

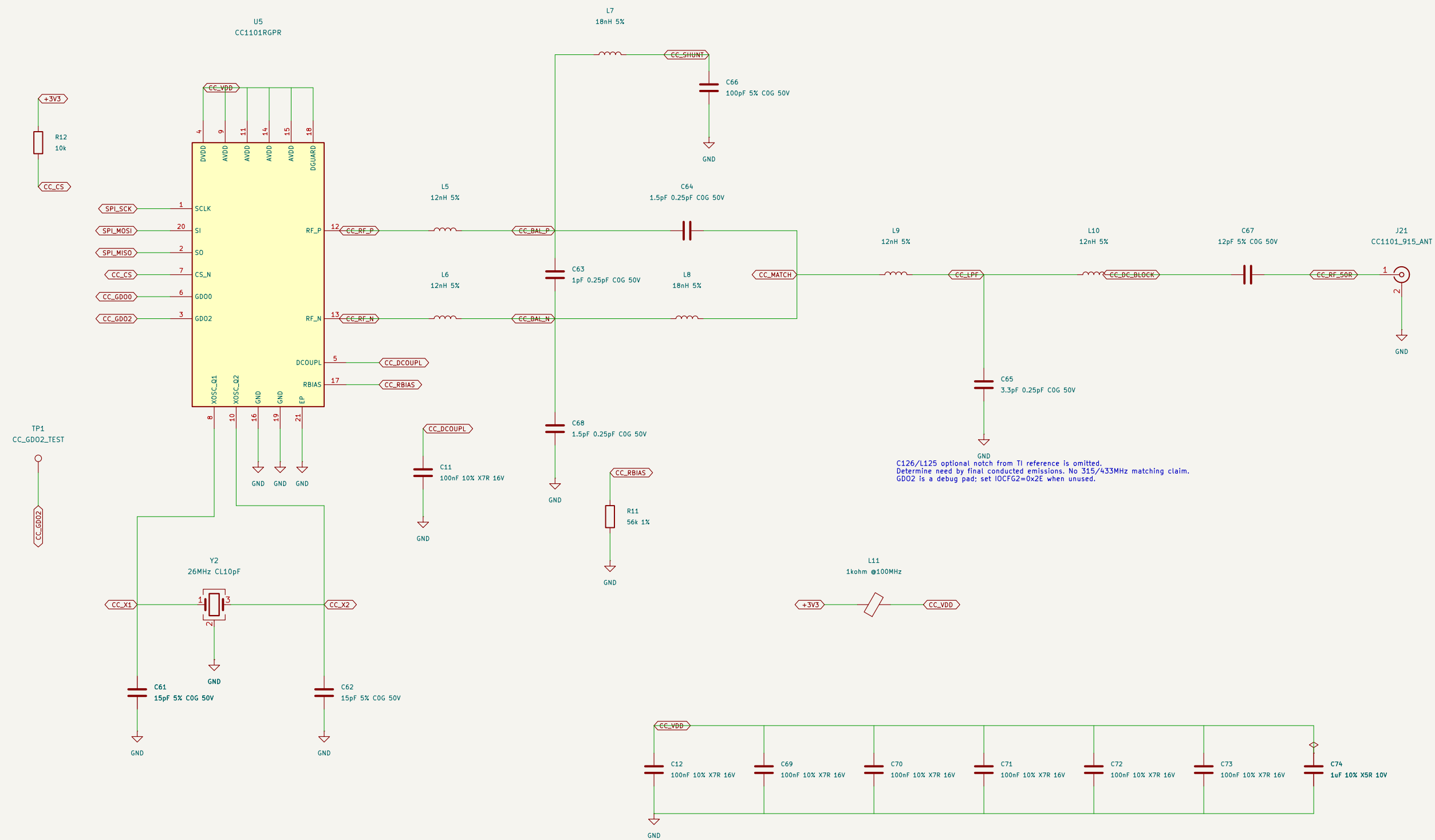
Title: SX1262 LoRa radio

Size: A3 Date: 2026-09-24
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Rev: A – engineering draft
Id: 12/23

CC1101 sub-GHz radio

868/915MHz frontend: TI SWRR045 rev3.0 / SWRS061I Fig11. RF layout and tuning are NOT qualified. Separate 50-ohm antenna.



Y2: ABM8-26.000MHZ-10-D-1-G-T
CL = 15pF / 2 + estimated 2.5pF stray = 10pF.
Measure oscillator frequency, startup and drive level.

CC_VDD flag: main +3V3 through passive L11; not an independent source.
Place one 100nF at each supply pin group, plus local 1uF bulk.

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Sheet: /CC1101 sub-GHz radio/
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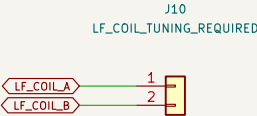
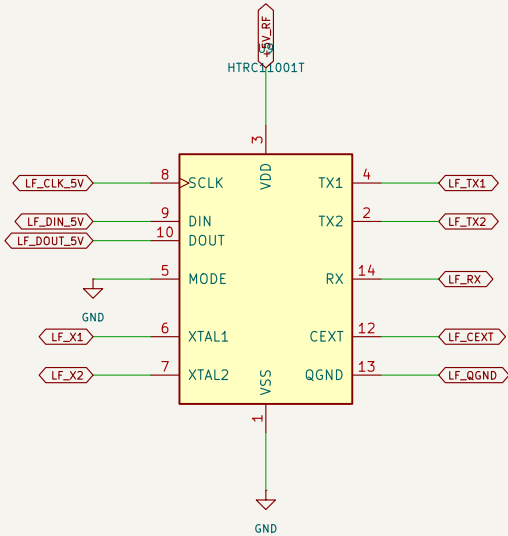
Title: CC1101 sub-GHz radio

Size: A3	Date: 2026-09-24
KiCad E.D.A. 10.0.6	

Rev: A – engineering draft
Id: 13/23

125 kHz reader analog circuit

HTRC110 requires its oscillator, QGND/CEXT bypass, resonant coil network and a timing-capable host interface.



Circuit review in progress – not released for fabrication

Sheet: /125 kHz reader analog circuit/
File: lf-radio.kicad_sch

Title: 125 kHz reader analog circuit

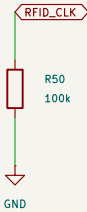
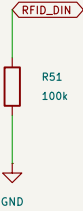
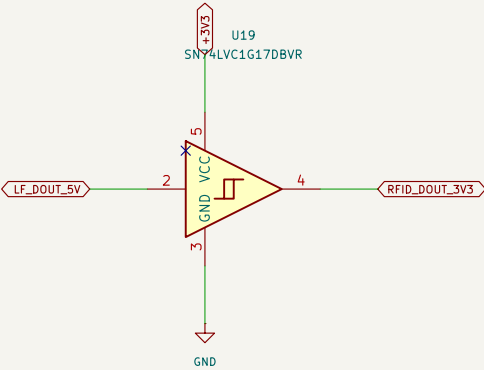
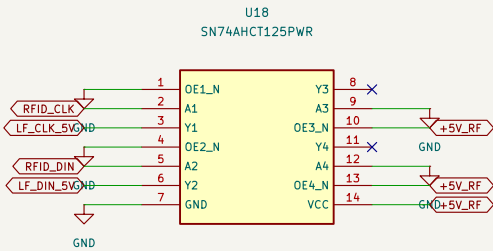
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KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft
Id: 14/23

LF reader logic translation

AHCT inputs accept 3.3 V logic at 5 V supply; LVC return buffer is powered at 3.3 V. Host timing remains a design requirement.



Circuit review in progress – not released for fabrication

Sheet: /LF reader logic translation/
File: lf-support.kicad_sch

Title: LF reader logic translation

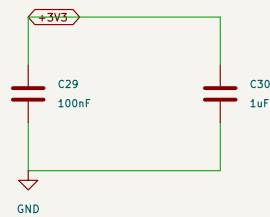
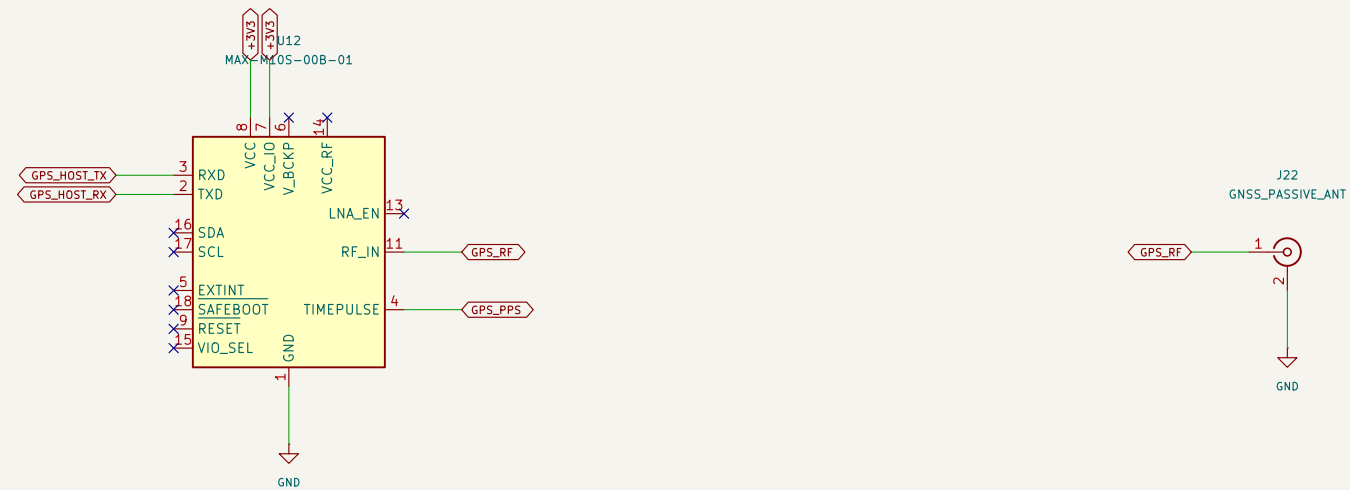
Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft
Id: 15/23

GNSS receiver and passive antenna

MAX-M10S passive antenna. V_BCKP remains open: cold starts after power removal. No direct LiPo connection.



Circuit review in progress – not released for fabrication

Sheet: /GNSS receiver and passive antenna/
File: gnss-antennas.kicad_sch

Title: GNSS receiver and passive antenna

Size: A3

Date: 2026-09-24

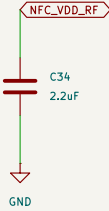
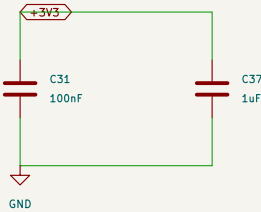
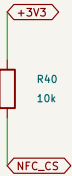
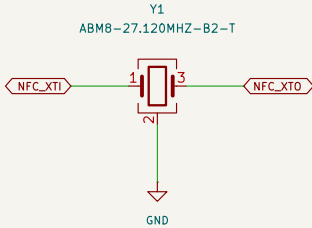
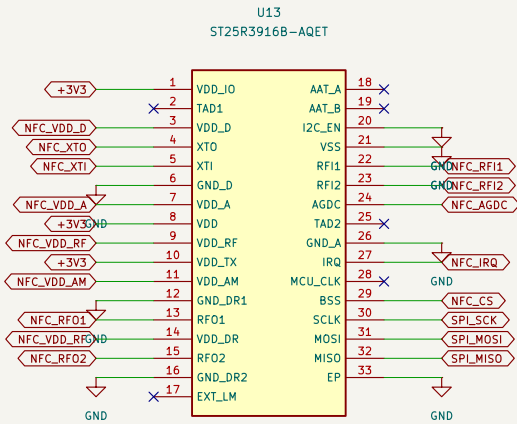
Rev: A – engineering draft

KiCad E.D.A. 10.0.6

Id: 16/23

NFC reader, oscillator and local supplies

ST25R3916B needs a tuned case-mounted loop and TX/RX matching network. Verify crystal load against the selected crystal.



Circuit review in progress – not released for fabrication

Sheet: /NFC reader, oscillator and local supplies/
File: nfc.kicad_sch

Title: NFC reader, oscillator and local supplies

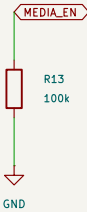
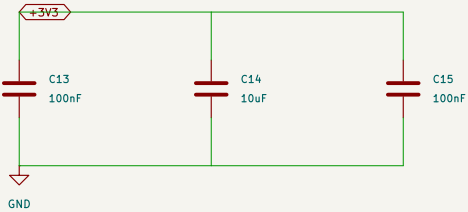
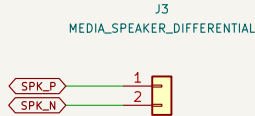
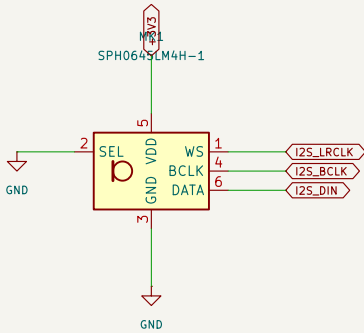
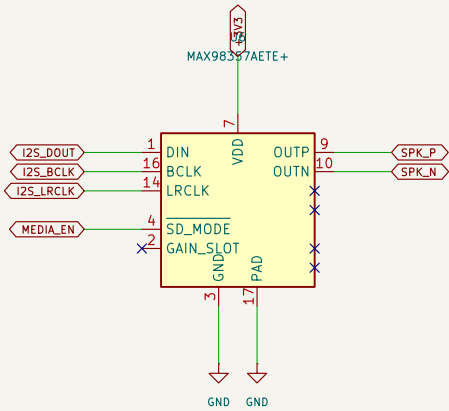
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KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft
Id: 17/23

Media speaker amplifier and digital microphone

Top-mounted media speaker uses differential outputs: neither speaker wire is ground. Digital microphone is case-bottom mounted.



Circuit review in progress – not released for fabrication

Sheet: /Media speaker amplifier and digital microphone/
File: audio.kicad_sch

Title: Media speaker amplifier and digital microphone

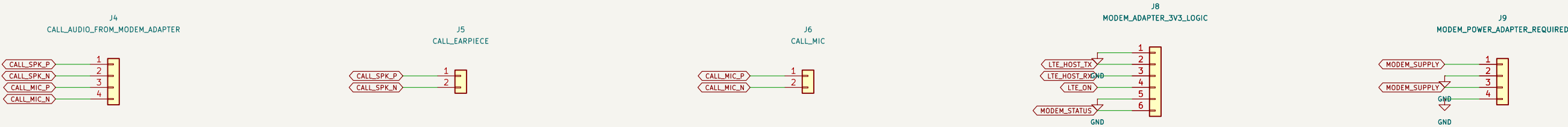
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KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft
Id: 18/23

Cellular adapter and call–audio harnesses

Replaceable modem adapter owns SIM, cellular RF and modem–level translation. No universal–carrier guarantee. Speaker top / mic bottom.



Circuit review in progress – not released for fabrication

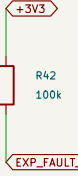
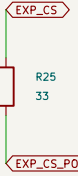
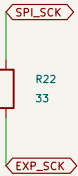
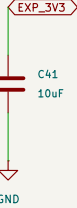
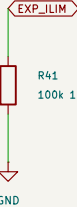
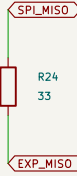
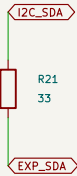
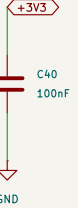
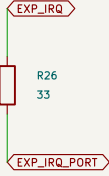
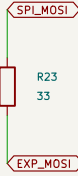
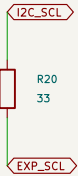
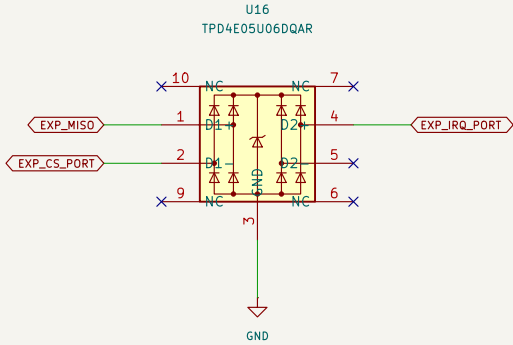
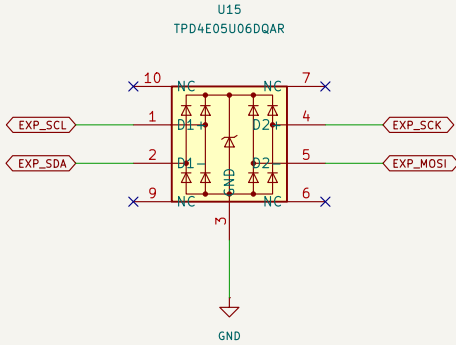
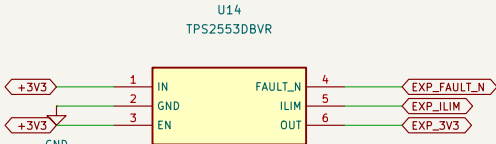
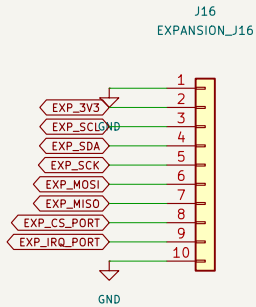
Sheet: /Cellular adapter and call–audio harnesses/
File: interfaces.kicad_sch

Title: Cellular adapter and call–audio harnesses

Size: A3	Date: 2026–09–24	Rev: A – engineering draft
KiCad E.D.A. 10.0.6		Id: 19/23

Protected external module port

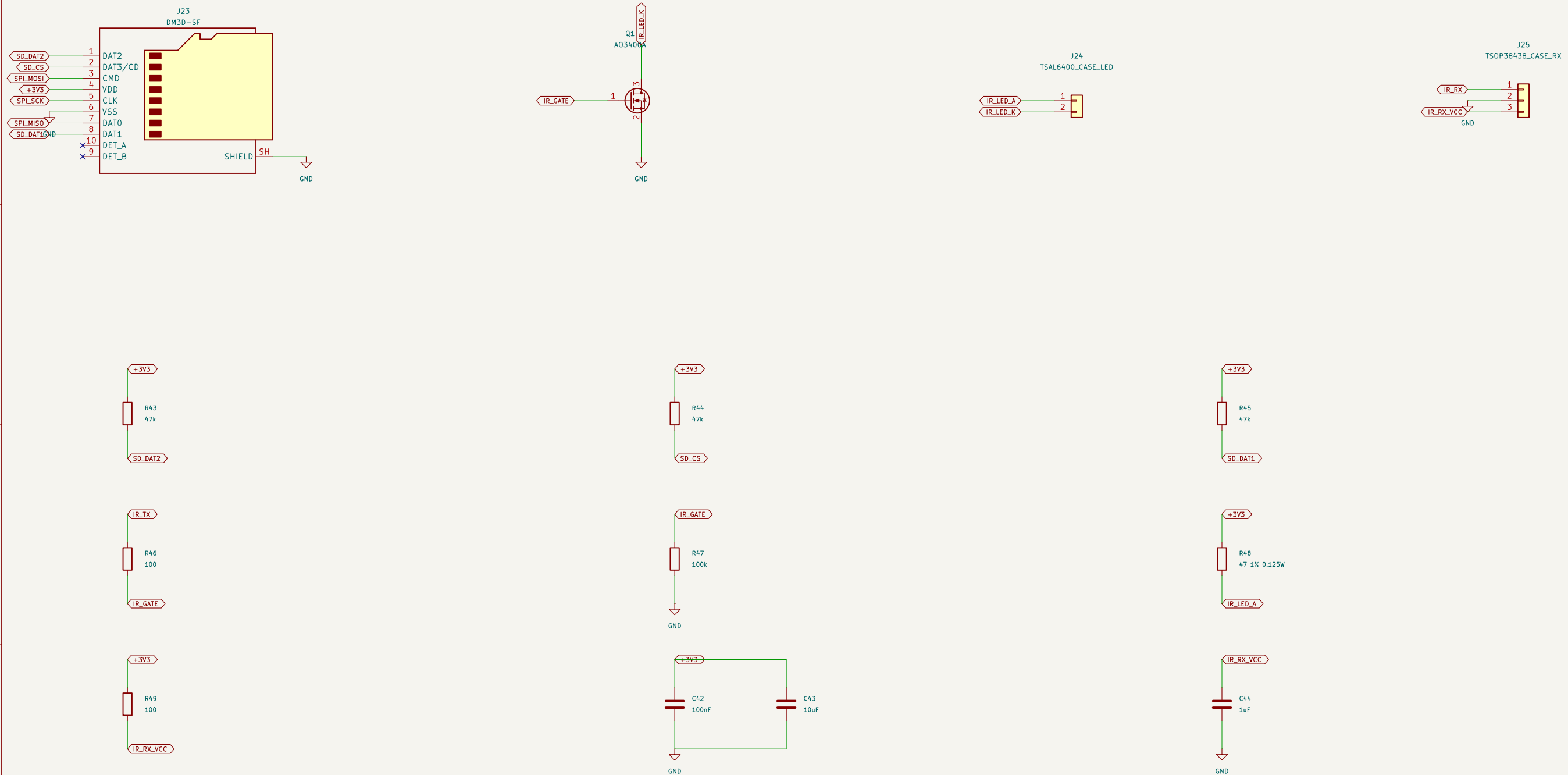
J16 pin order retained. Switch accessories with power off. TPS2553 limit and ESD path require qualification.



Circuit review in progress – not released for fabrication		
Sheet: /Protected external module port/ File: protection.kicad_sch		
Title: Protected external module port		
Size: A3	Date: 2026-09-24	Rev: A – engineering draft
KiCad E.D.A. 10.0.6		Id: 20/23

MicroSD and case-mounted infrared

SPI microSD, transistor-driven IR LED, filtered receiver supply. Confirm case LED/receiver polarity and GPIO3 boot behavior.



Circuit review in progress – not released for fabrication

Sheet: /MicroSD and case-mounted infrared/
File: storage-ir.kicad_sch

Title: MicroSD and case-mounted infrared

Size: A3

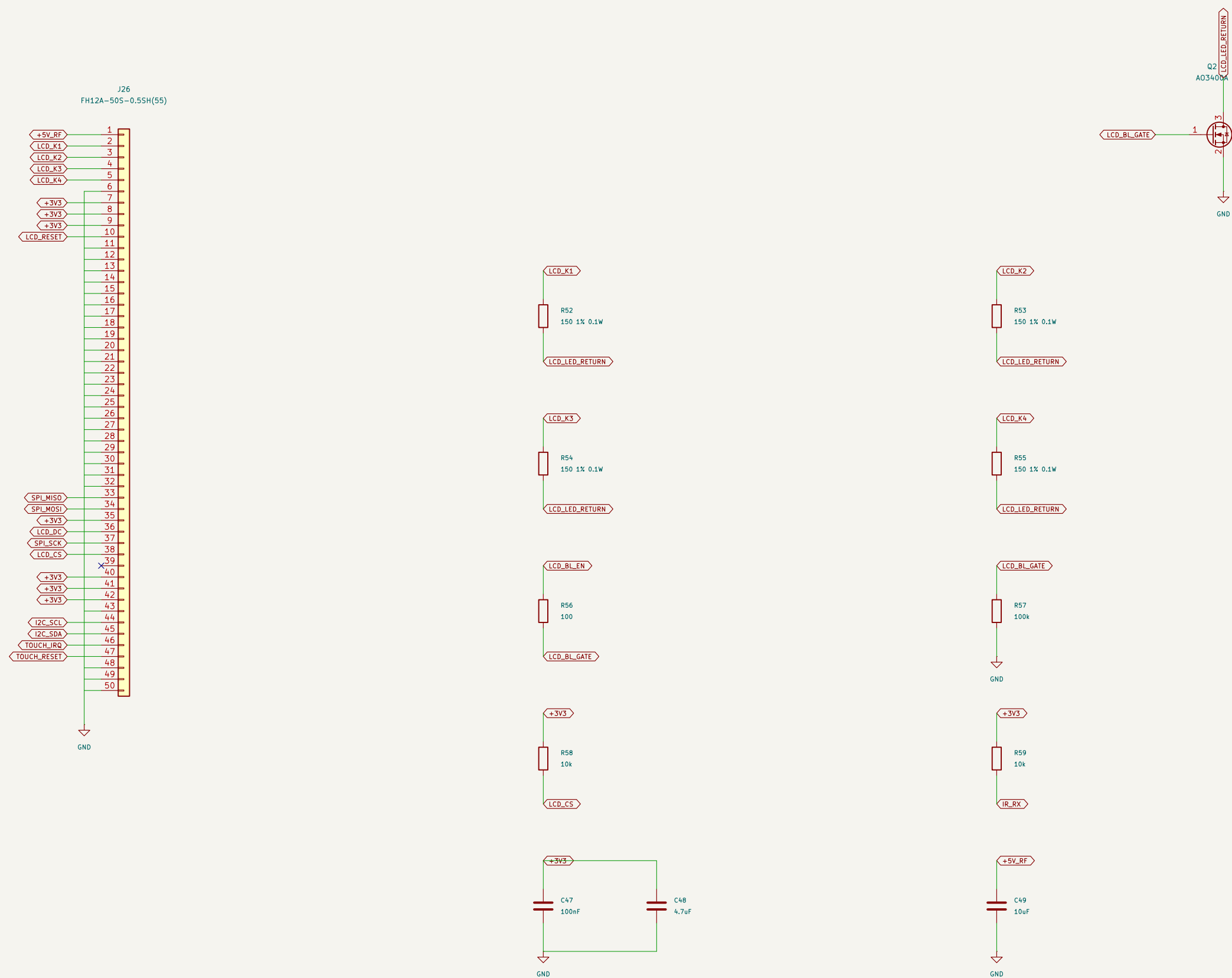
Date: 2026-09-24

Rev: A – engineering draft

Id: 21/23

External display, touch and backlight

ER-TFT024IPS-3 connector contract. Verify purchased CTP option, FPC orientation, LED branches and supply tolerances.



Circuit review in progress – not released for fabrication

Sheet: /External display, touch and backlight/
File: display.kicad_sch

Title: External display, touch and backlight

Size: A3

Date: 2026-09-24

Rev: A – engineering draft

Id: 22/23

Always-on charging supervisor

Separate supervisor selected. U3 remains the charger. Main host powers off at SW19. Circuit capture only; routing, STM32 firmware and fault qualification pending.

A

B

C

D

E

F

A

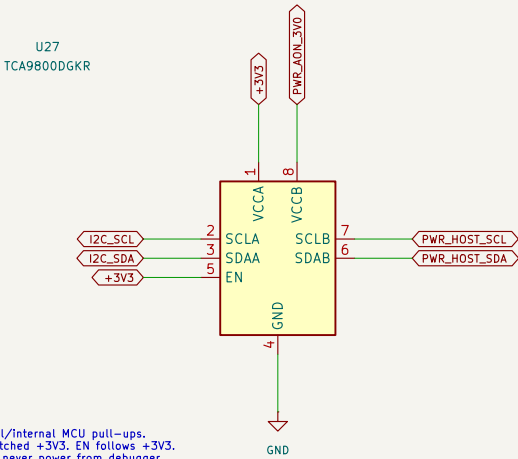
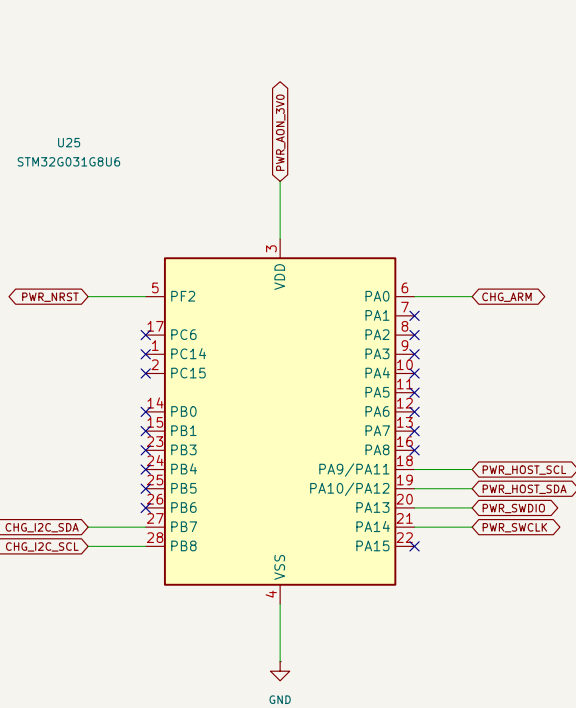
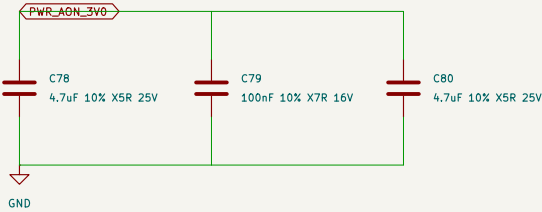
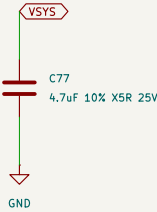
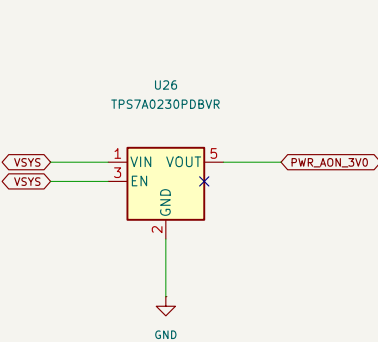
B

C

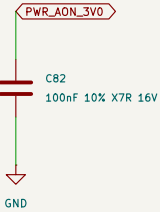
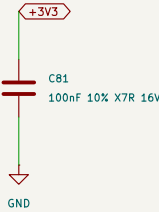
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E

F

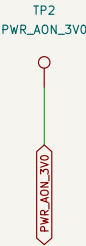
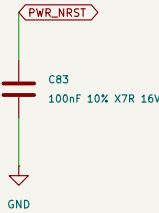
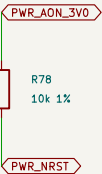
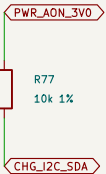
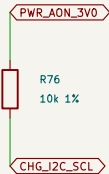


TCA9800 B side: NO external/Internal MCU pull-ups.
A-side pull-ups are on switched +3V3. EN follows +3V3.
Debug VTREF is sense only; never power from debugger.



I2C1: PB8/PB7 AF6 -> charger 0x6A + expander 0x41.
I2C2: remap PA9/PA10 to PA11/PA12, AF6.
Host slave address 0x42 reserved; protocol not implemented.

PA0: dedicated active-high charge arm. Reset analog/high-Z allows R79 at Q8 to hold charging disabled independently of U24. Never retain this output through standby or run bootloader while armed.



Circuit review in progress – not released for fabrication

Sheet: /Always-on charging supervisor/
File: power-supervisor.kicad_sch

Title: Always-on charging supervisor

Size: A3 Date: 2026-09-24
KiCad E.D.A. 10.0.6

Rev: A – engineering draft
Id: 23/23